Michael Ramirez

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Education

University of California, Santa Cruz — Santa Cruz, CA

B.S. in Computer Science | GPA: 3.82/4.0 — *Expected December 2027*

- **Coursework**: Machine Learning (Graduate), Artificial Intelligence, Data Structures & Algorithms, Probability Theory, Linear Algebra, Discrete Mathematics, Software Engineering, Object-Oriented Python, Computer Systems & Assembly Language

Experience

Group Tutor, CSE 13S (Computer Systems & C Programming) — Sep 2026 – Present

UC Santa Cruz — *Santa Cruz, CA*

- Lead group tutoring sessions for CSE 13S students on **C programming**, **memory management**, **pointers**, and **debugging**.
- Help students debug and complete assignments using **GDB**, **Valgrind**, and **Makefiles** in a **Linux** environment.
- Explain systems concepts including **bit manipulation**, **dynamic memory allocation**, and **data structures in C**.
- Guide students through tracking down **segfaults** and **memory leaks** by walking through their code line by line.

Undergraduate ML Researcher — Jan 2026 – Present

UC Santa Cruz — *Santa Cruz, CA*

- Reconstructed 3D scenes from **Meta ARIA** glasses data by modeling scene geometry with **3D Gaussian Splatting**.
- Enabled **open-vocabulary 3D scene search** by fusing **SAM2** segmentation masks with **vision-language embeddings**.
- Standardized training for **5+ lab researchers** by containerizing ML pipelines with **Docker** and **Kubernetes**.
- Presented research outcomes and pipeline progress to the **15-person lab** group and supervising professor.

United Way — Dec 2025 – Jun 2026

Software Engineer Intern

- Shipped **6+ frontend features** for a nonprofit's cross-platform **web and mobile app** as part of a **20-person team**.
- Implemented UI components in **React**, **React Native**, **TypeScript**, and **TailwindCSS** across web and mobile.
- Collaborated across dev and design teams, turning **Figma** mockups into working web and mobile screens.
- Delivered changes through **pull-request-based Git workflows** with peer code review, tracking tasks in **Linear**.

Projects

Visage | *TypeScript, Bun, Python, FastAPI, PostgreSQL/pgvector, InsightFace, Gemini 2.5* — Feb 2026 – Present

- Built **voice commands** on **MentraOS** smart glasses that let users save, recognize, look up, and forget people they've met.
- Linked faces to conversation context by pairing **InsightFace** encodings with **Gemini 2.5 Flash** transcript extraction.
- Matched returning faces per user with **pgvector** L2-distance search over **512-d** face embeddings in a **FastAPI** backend.
- Cut **photo capture latency** by triggering the glasses camera on **partial speech transcripts** before commands finalized.
- Filtered **false face detections** using **confidence thresholds**, minimum size checks, and **upscaling** of small glasses images.

HardwareHub | *TypeScript, Hono, Supabase, Auth0, React, Opennote API* — Jan 2026

- Designed an **ESP32/Arduino** learning platform backend with **39 REST endpoints** in **Hono** over **6 Supabase tables**.
- Built a **KiCad-style board editor** where users drag **17 hardware modules** onto a canvas and wire inputs to outputs.
- Personalized studying by generating quizzes from lesson content and student notes via the **Opennote Practice API**.
- Secured user accounts and API routes with **Auth0** passwordless email login and **JWT verification** middleware in Hono.
- Motivated learners with **17 unlockable achievements** for completing lessons, acing quizzes, and maintaining streaks.

Leadership

Research Lab Lead & Instruction Officer — Oct 2026 – Present

Association for Computing Machinery (ACM), UC Santa Cruz — *Santa Cruz, CA*

- Lead ACM's undergraduate research lab, mentoring student teams through **ML and systems research** projects.
- Run **technical workshops** and instructional sessions for ACM members on tools like **PyTorch**, **Git**, and **Docker**.
- Help members scope research ideas into **quarter-long projects** and present results at **end-of-quarter showcases**.

Awards & Health/Wellness Chair | *Kappa Sigma, UC Santa Cruz* — Oct 2026 – Present

- Organize **health and wellness events** for chapter members, promoting physical fitness and mental health resources.
- Recognize member achievements in **academics**, **athletics**, and **community service** through chapter awards.

Awards & Honors

Best Hardware Hack – CruzHacks 2026 (UCSC) | **3rd Place** – Reboot the Earth 2025 (United Nations hackathon)

Technical Skills

Languages: Python, TypeScript/JavaScript, C, C++, SQL (PostgreSQL), RISC-V Assembly, Bash
Frameworks & Libraries: PyTorch, FastAPI, React, React Native, TailwindCSS, Bun, Vite, Hono, LangChain/LangGraph
Developer Tools: Docker, Kubernetes, Git, Linux, PostgreSQL/pgvector, Supabase
Concepts: REST APIs, vector search & embeddings, computer vision, LLM agent workflows